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Defining and Estimating a Trading Area

DAVID L. HUFF

What are the conceptual properties of a trading area?
What is the definition of the term?
What testable propositions are currently available to validate its properties and thus give precision to the definition?
This article gives answers to these questions.

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MARKET analysts have long speculated about the nature and scope of trading areas. Such speculations have been based primarily upon conclusions drawn from empirical studies.

However, except for the "gravitationalists," few analysts, if any, have formulated their conclusions into propositions that are capable of being verified or refuted by empirical test. As a consequence, the conceptual properties of a trading area are extremely vague and perhaps in error. Furthermore, existing techniques for estimating trading areas are limited and subject to question.

The objectives of the present article are threefold: (1) to appraise the principal techniques used to delineate retail trading areas; (2) to enumerate significant conclusions derived from empirical studies using such techniques; and (3) to advance an alternative technique, believed to be better conceptually and superior predictively.

Estimating Techniques

The methods employed to delineate trading areas, particularly retail trading areas, generally involve surveys or the use of empirically derived mathematical formulations.

Survey Techniques

Typically, in the case of survey techniques, a sample of individuals representing either households or firms are interviewed at their places of origin or at the particular firm or center for which the trading area is being estimated.

Such interviews are designed primarily to determine the kind or kinds of products that are purchased by each respondent, the frequency of patronage, and the home-base location of each respondent. These data can then be used to prepare a map from which inferences can be drawn concerning the nature and scope of the trading area.

As a result of trading area studies using survey techniques, a number of important empirical regularities have been shown to exist:

1. The proportion of consumers patronizing a given shopping area varies with distance from the shopping area.
2. The proportion of consumers patronizing various shopping areas varies with the breadth and depth of merchandise offered by each shopping area.
3. The distances that consumers travel to various shopping areas vary for different types of product purchases.
4. The "pull" of any given shopping area is influenced by the proximity of competing shopping areas.

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A number of market analysts have attempted to generalize about the nature and scope of trading areas by citing *specific* conclusions drawn from such empirical studies. For example, it is often maintained that the trading area for a certain size of retail facility offering a particular class of products will encompass a radial distance of some specified number of miles, of which the primary trading area will involve a certain proportion of the total area, etc., etc.

But generalizations of this kind may be subject to a great deal of error because of differences among regions with respect to transportation facilities, topographical features, population density, and the locations of competing firms.

Mathematical Techniques

A few analysts have attempted to formalize some of the general conclusions drawn from empirical studies. They have expressed their ideas in terms of mathematical propositions that are capable of being tested empirically. The work that has been done in this area is limited primarily to the so-called "retail gravitationalists."

Notable among these is William J. Reilly who made a significant contribution by formalizing a number of empirical observations concerning consumer shopping movements between cities.[1] The nature of his formal construct is shown below:

$$\text{(1)} \qquad \frac{B_a}{B_b} = \left(\frac{P_a}{P_b}\right)\left(\frac{D_b}{D_a}\right)^2$$

where B_a = the proportion of the retail business from an intermediate town attracted by city A;
B_b = the proportion of the retail business from an intermediate town attracted by city B;
P_a = the population of city A;
P_b = the population of city B;
D_a = the distance from the intermediate town to city A; and,
D_b = the distance from the intermediate town to city B.

The extensive empirical tests of Reilly's model that were made by P. D. Converse are also noteworthy.[2] In addition, Converse is to be credited for making a significant modification of Reilly's original formula.[3] This modification made it possible to calculate the approximate point between two competing cities where the trading influence of each was equal. As a consequence, a city's retail trading area could be delineated by simply calculating and connecting the breaking points between it and each of the competing cities in the region (see Figure 1). The breaking point formula derived by Converse is:[4]

$$\text{(2)} \qquad D_b = \frac{D_{ab}}{1 + \sqrt{\frac{P_a}{P_b}}}$$

where D_b = the breaking point between city A and city B in miles from B;
D_{ab} = the distance separating city A from city B;
P_b = the population of city B; and
P_a = the population of city A.

Limitations. The significance of the pioneering efforts of both Reilly and Converse to provide a systematic basis for estimating retail trading areas cannot be denied. The variables employed, the functional relationships advanced, and the estimated parameters provide precise and meaningful hypotheses that can be tested empirically.

However, there are several important conceptual and operational limitations associated with the use of the "Reilly-type" model.

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1 William J. Reilly, *Methods for Study of Retail Relationships* (Austin: The University of Texas, Bureau of Business Research, Research Monograph, No. 4, 1929).

2 P. D. Converse, *A Study of Retail Trade Areas in East Central Illinois* (Urbana: University of Illinois, Bureau of Economic and Business Research, Business Studies, No. 2, 1943); and *Consumer Buying Habits in Selected South Central Illinois Communities* (Urbana: University of Illinois, Bureau of Economic and Business Research, Business Studies, No. 6, 1948).

3 This modification as well as other changes that Converse made of Reilly's original model is well summarized in P. D. Converse, "New Laws of Retail Gravitation," JOURNAL OF MARKETING, Vol. 14 (January, 1949), pp. 379-384.

4 Converse did not demonstrate how he derived the breaking point formula from Reilly's equation. However, the proof of such a derivation is simply:

(i) $\frac{B_a}{B_b} = 1$

(ii) $\left(\frac{P_a}{P_b}\right)\left(\frac{D_b}{D_a}\right)^2 = 1$

(iii) $\frac{D_b}{D_a} = \sqrt{\frac{P_b}{P_a}}$

(iv) $D_a = D_{ab} - D_b$

(v) $\frac{D_b}{D_{ab} - D_b} = \sqrt{\frac{P_b}{P_a}}$

(vi) $\frac{D_{ab}}{D_b} - 1 = \sqrt{\frac{P_a}{P_b}}$

(vii) $\frac{D_{ab}}{1 + \sqrt{\frac{P_a}{P_b}}} = D_b$

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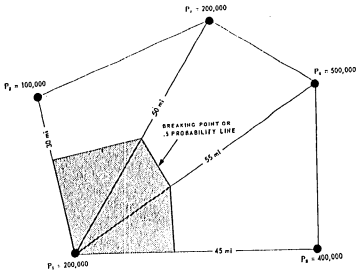

Calculation of Breaking Points:

$$D_2 = \frac{D_{12}}{1 + \sqrt{\frac{P_1}{P_2}}} = \frac{30}{1 + \sqrt{\frac{200,000}{100,000}}} = 12.4$$

$$D_3 = \frac{D_{13}}{1 + \sqrt{\frac{P_1}{P_3}}} = \frac{50}{1 + \sqrt{\frac{200,000}{200,000}}} = 25.0$$

$$D_4 = \frac{D_{14}}{1 + \sqrt{\frac{P_1}{P_4}}} = \frac{55}{1 + \sqrt{\frac{200,000}{500,000}}} = 33.0$$

$$D_5 = \frac{D_{15}}{1 + \sqrt{\frac{P_1}{P_5}}} = \frac{45}{1 + \sqrt{\frac{200,000}{400,000}}} = 26.3$$

FIGURE 1. Estimating a trading area with the breaking point formula.

First, the breaking point formula, as it now exists, is incapable of providing graduated estimates above or below the break-even position between two competing centers. See Figure 1. As a consequence, it is impossible to calculate objectively the *total* demand for the product(s) or service(s) of a particular distribution center.

Second, when the breaking point formula is used to delineate retail trading areas of several shopping areas within a given geographical area, the overlapping boundaries that result are inconsistent with the basic objective of the formula's use; to calculate the boundaries between competing shopping areas where the competitive position of each is equal. Furthermore, in the case of multi-trading area delineations derived from using the breaking point formula, there may be areas that are not even within the confines of any shopping area's trading area. Such a development is certainly not very realistic. A visual exemplification of these conditions is shown in Figure 2.

Finally, the parameter which was originally estimated empirically by Reilly should not be interpreted as a constant for all types of shopping trips as so many analysts have assumed. It seems quite logical to hypothesize that such an exponent will vary, depending on the type of shopping trip involved. As a result, a distribution center may have several different trading areas corresponding to the different classes of products that it sells.

An Alternative Model

An alternative model will now be presented which overcomes the limitations described above.

The principal focus of the model is on the consumer rather than on the firm. It is, after all, the consumer who is the primary agent affecting the trading area of the firm. The model describes the process by which consumers choose from among acceptable alternatives, a particular distribution center (a firm or group of firms) to obtain specific goods and services. A formal expression of the model is:

$$P_{ij} = \frac{\dfrac{S_j}{T_{ij}^{\lambda}}}{\sum_{j=1}^{n} \dfrac{S_j}{T_{ij}^{\lambda}}} \qquad (3)$$

where P_{ij} = the probability of a consumer at a given point of origin i traveling to a particular shopping center j;

S_j = the size of a shopping center j (measured in terms of the square footage of selling area devoted to the sale of a particular class of goods);

T_{ij} = the travel time involved in getting from a consumer's travel base i to a given shopping center j; and

λ = a parameter which is to be estimated empirically to reflect the effect of travel time on various kinds of shopping trips.

The *expected* number of consumers at a given place of origin i that shop at a particular shopping center j is equal to the number of consumers at i multiplied by the probability that a consumer at i will select j for shopping. That is,

$$E_{ij} = P_{ij} \cdot C_i \qquad (4)$$

where E_{ij} = the expected number of consumers at i that are likely to travel to shopping center j; and

C_i = the number of consumers at i.

In many respects the preceding model resembles the original model formulated by Reilly. It differs, however, in several important respects.

First, the alternative model is not merely an empirically contrived formulation. It represents a theoretical abstraction of consumer spatial behavior. As a result, mathematical conclusions can be deduced from the model which, in turn, can be inter-

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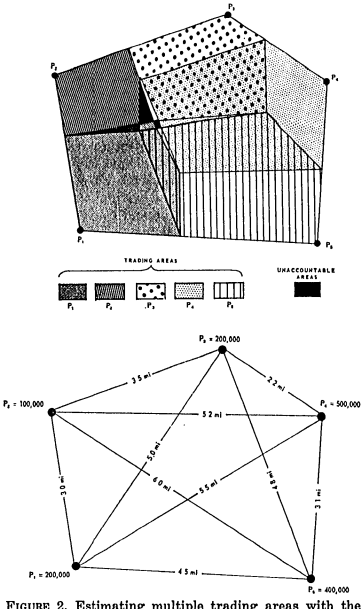

FIGURE 2. Estimating multiple trading areas with the breaking point formula.

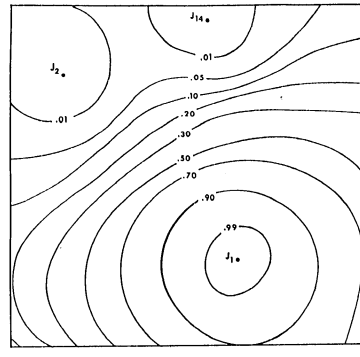

FIGURE 3. A retail trading area portrayed in terms of probability contours. Source: David L. Huff, *Determination of Intra-urban Retail Trade Areas* (Los Angeles: University of California, Real Estate Research Program, 1962).

preted in terms of their behavioral implications.[5]

Second, the alternative model estimates the likelihood of a consumer (P_{ij}) or the number of consumers (E_{ij}) patronizing a particular shopping area by taking into consideration *all* potential shopping areas simultaneously.

Third, the parameter λ is not assumed to be to the second power. Rather, it is assumed to vary with different types of product classes. For example, in an initial pilot study λ was found to be 2.723 for shopping trips involving furniture and 3.191 for trips involving clothing purchases.[6] The respective magnitudes of these estimates simply reflect the comparative amounts of time that consumers are willing to expend for each of these two product classes. The larger the estimated value of λ, the smaller will be the time expenditure. Similarly, the larger the estimated value of λ, the more restrictive will be the scope of the trading area.

Finally, equations (3) and (4) enable a retail trading area to be graduated in terms of demand gradients. These gradients are expressed as probability contours ranging from $P < 1$ to $P > 0$. An illustration of how these contours look when mapped is illustrated in Figure 3, in which a partial retail trading area has been calculated for shopping center J_1.

If the retail trading areas of shopping centers J_2 and J_{14} had also been calculated and superimposed over the trading area of J_1, it would be seen that parts of each shopping center's trading area envelop parts of the others. Furthermore, where these envelopments occur at intersections of contours having the same probability values, it would be possible to determine the breaking points between each of these competing centers.

General Conclusions

The following general conclusions can now be drawn concerning the nature and scope of a trading area:

1. A trading area represents a *demand surface* containing potential customers for a specific product(s) or service(s) of a particular distribution center.

2. A distribution center may be a *single firm or an agglomeration of firms.*

3. A demand surface consists of a series of *demand gradients* or zones, reflecting varying customer-sales potentials. An exception to the condi-

[5] For a discussion of the theoretical aspects of the model, see David L. Huff, "A Probabilistic Analysis of Consumer Spatial Behavior," in William S. Decker, Editor, *Emerging Concepts in Marketing* (Chicago: American Marketing Association, 1963), pp. 444-450.

[6] David L. Huff, *Determination of Intra-urban Retail Trade Areas* (Los Angeles: University of California, Real Estate Research Program, 1962).

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tion of demand gradients would be in the rare case in which only one distribution center existed in a unique geographical setting, thus representing an absolute monopoly in providing products and/or services that are of an absolute necessity. Under these conditions, no gradients would exist but rather a single homogeneous demand plane.

4. Demand gradients are of a *probabilistic* nature, ranging from a probability value of less than one to a value greater than zero (except in the complete monopoly situation in which the probability value equals one).

5. The total potential customers encompassed within a distribution center's demand surface (trading area) is *the sum of the expected number of consumers from each of the demand gradients.*

6. Demand gradients of competing firms overlap; and where gradients of like probability intersect, *a spatial competitive equilibrium* position is reached.

Final Definition

Accordingly, a trading area can be defined as: *A geographically delineated region, containing potential customers for whom there exists a probability greater than zero of their purchasing a given class of products or services offered for sale by a particular firm or by a particular agglomeration of firms.*

This definition can be expressed symbolically as:

$$T_j = \sum_{i=1}^{n} (P_{ij} \cdot C_i) \qquad (5)$$

where T_j = the trading area of a particular firm or agglomeration of firms j, that is, the total expected number of consumers within a given region who are likely to patronize j for a specific class of products or services;

P_{ij} = the probability of an individual consumer residing within a given gradient i shopping at j; and

C_i = the number of consumers residing within a given gradient i.

By comparison, the currently accepted definition of the term "trading area," as expressed by the Committee on Definitions of the American Marketing Association, is: "A district whose size is usually determined by the boundaries within which it is economical in terms of volume and cost for a marketing unit to sell and/or deliver a good or service."[7] This definition provides little insight concerning the nature and scope of a trading area. Furthermore, this definition implies that a trading area does not encompass the entire region within which potential demand exists, but rather only that portion which a marketing unit finds it economical to sell and/or deliver a good or service.

It is obvious, however, that in order for a marketing unit to determine the specific region that it finds economical for distribution purposes, it first has to assess the demand in the entire potential trading area. In addition, no matter what cost variable is considered, for example, delivery or promotion, it is very likely that the cost of the service under consideration will not turn out to be a very satisfactory determinant of any precisely bounded trading area as suggested by the Committee's definition. Finally, this definition conveys the image that it is the marketing unit that determines the trading area rather than the consumer.

[7] Committee on Definitions of the American Marketing Association, *Marketing Definitions: A Glossary of Marketing Terms* (Chicago: American Marketing Association, 1960).

MARKETING MEMO

The Problem Of Profit Management Starts Early . . .

. . . A store's profitability is heavily influenced by what happens before it opens. Once a distributor has decided where to put a store, and how big to build it, he has pretty well set at least the range of its future profits.

—"The Economics of Food Distributors," McKinsey-General Foods Study, published by General Foods Corporation, October, 1963.

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